Preliminary matched-V_2/readout-decay experiment

Seed 0; 8 ReLU scales × 8 decay values = 64 completed cells

Question and design. This experiment repeats the unfrozen accessibility sweep while applying the same phase-2 weight decay λ to V_2 and to the $d4$ readout row $W[2]$. The purpose is to remove the V2-only experiment's scale escape, in which V_2 shrank while the unpenalized readout grew. The preliminary grid covers $\alpha \in \{0.1, 0.3, 0.5, 0.75, 1, 1.5, 2.5, 4\}$ and $\lambda \in \{0, 10^{-4}, 3{\cdot}10^{-4}, 10^{-3}, 3{\cdot}10^{-3}, 10^{-2}, 3{\cdot}10^{-2}, 10^{-1}\}$.

Is R_0 controlled by the ReLU coefficient? Yes. At $d4$ onset the pretrained first-block matrix is replaced by $V_1 \mapsto \alpha V_1$. For $\alpha > 0$, ReLU homogeneity gives $\mathrm{ReLU}(h_0(\alpha V_1)^\top) = \alpha\,\mathrm{ReLU}(h_0 V_1^\top)$, so α directly controls the first-block features and therefore the measured accessibility $R_0 = Q_2/Q_1$. The relation is monotone but not linear because the residual term h_0 is not rescaled. In seed 0, each α gave exactly one R_0 across all eight λ values:

α	0.10	0.30	0.50	0.75	1.00	1.50	2.50	4.00
R_0	.038	.113	.181	.256	.319	.411	.510	.568

Thus the theoretical boundary $R_c = 1/\sqrt{8} = 0.354$ is crossed between $\alpha = 1$ and $\alpha = 1.5$. Across seeds, α remains the controlled intervention and R_0 the measured response; small seed-to-seed shifts are expected.

Preliminary result. Matched readout decay does *not* sharpen a compositional transition near R_c in seed 0. The largest compositional shares occur only under very weak decay: for the supercritical $\alpha = 2.5$ cell, f_{2+2} rises from 0.085 at $\lambda = 0$ to 0.114 at $3{\cdot}10^{-4}$, then collapses to 0.017 at 10^{-3} and approximately zero by $3{\cdot}10^{-3}$. Averaged across α, f_{2+2} falls from 0.043 at $\lambda = 0$ to 0.008 at 10^{-3}, 0.002 at 10^{-2}, and 0.0005 at 10^{-1}.

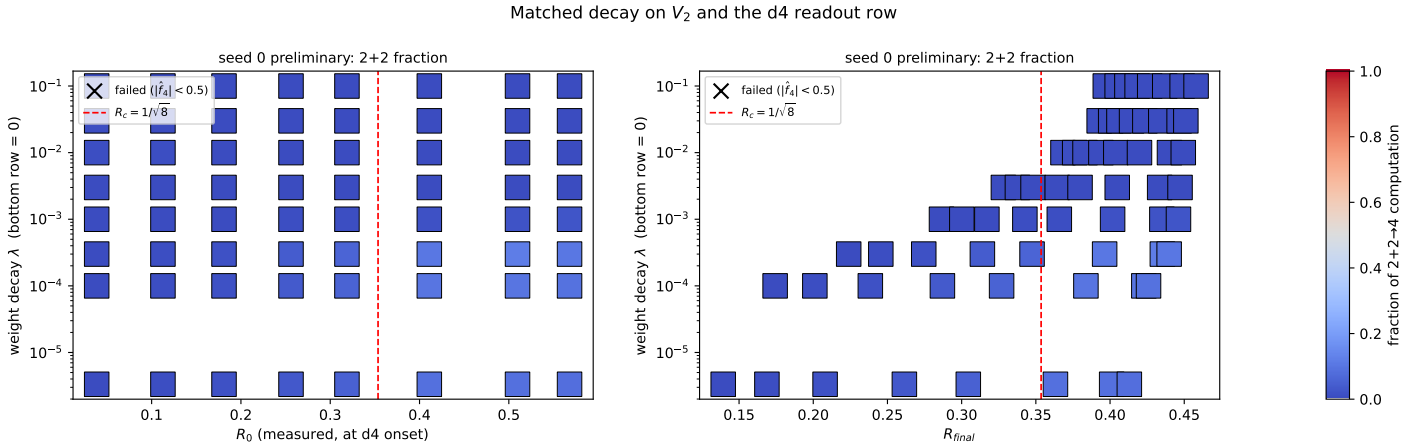


Figure 1: Direct seed-0 analogue of Fig. 1 in the V2-only report. Color is the weighted $2{+}2 \to 4$ fraction; the dashed line is R_c. Despite upward/rightward drift in R_{final} under decay, cells remain blue: becoming accessible does not produce a compositional phase when the $d4$ readout is decayed together with V_2.

Interpretation. The scale escape is removed—both V_2 and its $d4$ readout are penalized—but the network responds by suppressing the measured block-2 compositional route rather than balancing it. At the same time, $R(t)$ generally drifts upward (for example, at $\alpha = 0.1$, $R_{\mathrm{final}} = 0.139$ at $\lambda = 0$ and 0.368 at 10^{-2}). The coexistence of high R_{final}, near-zero f_{2+2}, and solved $d4$ is consistent with the first-block $d4$ leak/residual shortcut taking over. This is a strong preliminary pattern, but uncertainty and cross-seed consistency must wait for the full 192-cell merge.